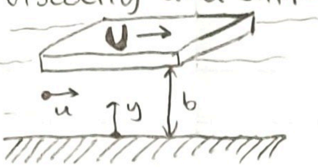
ME 106 MT1 Cheat Sheet

LEC1 → matching units.
what is a fluid? : a substance that DEFORMS under the action of an applied shear stress
Key: The fluid deforms CONTINUOUSLY!

Properties of fluids:
① Density (kg/m³)
② SG (unitless)
③ VISCOCITY! → resistance to flow under an applied shear stress.

These 3 properties define all NEWTONIAN fluids.

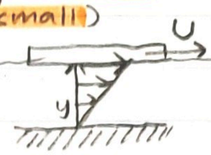
Viscocity and slippage: ① NO-SLIP boundary condition
- fluid particles DO NOT SLIP against a wall boundary.



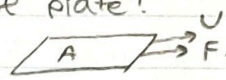
implies ⟹ $u(y=0) = 0$ // no velocity of fixed surface
$u(y=b) = U$ // matches velocity of plate!

② Linear velocity variance (when b is small)

$$\boxed{u(y) = U\frac{y}{b}}$$



③ $U \propto \frac{F_t * b}{A}$



Recall $F_t/A = \tau$ (shear stress)

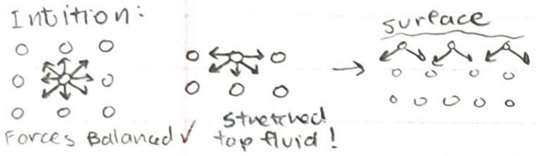
$\tau \propto U/b \Rightarrow \tau = \mu\, U/b = \mu \frac{dU}{dy}$

μ → coefficient of viscocity

SURFACE TENSION

cohesive vs adhesive → between 2 same particles vs between 2 different particles

Intuition:



$F \propto \ell$ $\quad F = \sigma \ell$ where σ = coefficient of surface tension

Examples: water droplet

$P_{inside} > P_{outside}$!

$\underbrace{P_i * \pi R^{\cancel{2}}}_{\text{pressure F}} = \underbrace{\sigma * 2\pi \cancel{R}}_{\text{surf ten}}$

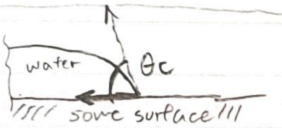
$\boxed{P_i = \frac{2\sigma}{R}}$

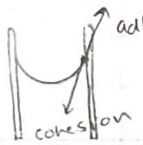
bubble (Air, water, air)

$P_i * \pi R^2 = (\sigma * 2\pi R) * 2$

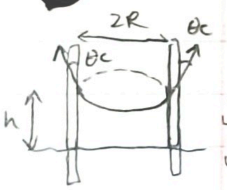
$\boxed{P_i = \frac{4\sigma}{R}}$

θ_c → contact angle





θ_c:
- 0 perfect wetting
- $0 \le \theta \le 90°$ high wetting, concave meniscus
- $90° \le \theta \le 180°$ low wetting, convex meniscus
- 180° perfectly non-wetting

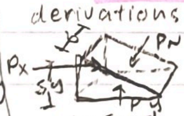


$\underbrace{\rho g h \pi R^2}_{\text{weight of water beneath}} = \underbrace{\sigma\, 2\pi R \cos\theta_c}_{\text{surface tension forces y-component}}$

$$\boxed{h = \frac{2\sigma\cos\theta_c}{\rho R g}}$$

FLUID STATICS

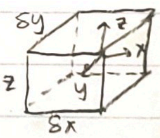
Rule: pressure is independent of direction in a fluid!
derivations typically always start with $F = ma_x$ crucial start pt.
$= \rho\,\delta V a_x = \rho\,\delta x \delta y \delta z\, a_x$

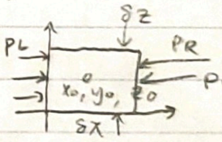


$P_x - P_N = \frac{1}{2}\rho\,\delta x\, a_x$

$\lim_{\delta x \to 0}(P_x - P_N) = 0 \quad P_x \to P_N$!

Accelerating Fluids





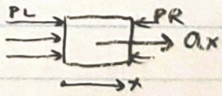
$P_L(\delta z \delta y) - P_R(\delta z \delta y) = \rho(\delta x \delta y \delta z) a_x$

Taylor expansion proof → $-\frac{\partial P}{\partial x} = \rho a_x$

$-\frac{\partial P}{\partial y} = \rho a_y$

$-\frac{\partial P}{\partial z} = \rho(a_z + g)$

Intuition?



$P_L > P_R$

$\frac{\partial P}{\partial x}$ is negative if a_x is positive!

Pressure fields $P(x,y,z,t)$

$\vec{\nabla}\phi = \begin{Bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{Bmatrix}\phi$

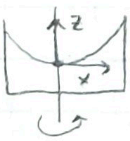
del operator!

$$-\vec{\nabla}P = \rho\vec{a} + \rho g\hat{k} \;!$$

$$-\begin{Bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{Bmatrix} P = \rho\begin{Bmatrix} a_x \\ a_y \\ a_z \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \rho g \end{Bmatrix}$$

match!

Aside: Parabolic surface of rotating fluid



$a = \omega^2 x\,\hat{x}$

$\frac{\partial z}{\partial x} = \frac{\partial P/\partial x}{\partial P/\partial z} = \frac{-\rho a_x}{-\rho g}$

$= \frac{\omega^2 x}{g}$

Parabola!

$\int \delta z = \int \frac{\omega^2 x}{g}\delta x \rightarrow \boxed{z = \frac{\omega^2}{g}\frac{x^2}{2} + C}$

BUOYANCY $F_z = \rho g V$

① intuitive interpretation
- repl. object with a body of water same volume. Said body must suspend there

② Mathy justification

$d\vec{F} = -P\,d\vec{s}$ where $|d\vec{s}| = dA$

$dF_z = -(-\rho g z)\,d\vec{s}\,\hat{k}$

$F_z = \rho g \int_A z\,d\vec{s}\,\hat{k}$

Divergence Theorem! $\int_A G \cdot d\vec{s} = \int_V (\vec{\nabla}\cdot G)\,dV$

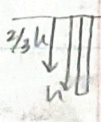
$F_z = \rho g \int_V \vec{\nabla}\cdot z\hat{k}\,dV \quad (\vec{\nabla}\cdot z\hat{k} = 1)$!

$= \rho g \int_V 1\,dV = \rho g V$ ✓

Net Forces acting on underwater object

$F = \int \rho g z\,dA = \rho g z_c$ // this is the magnitude

$z_{effective} = \frac{\int z^2 dA}{\int z\,dA}$ → Area moment of inertia / → centroid of the surface



for rectangles the value is 2/3 h!

FLUID KINEMATICS

- Eulerian : state function of time and space.
  vs ↳ better for fluids
- Lagrangian: follows a specific particle at times t.
  ↳ better for solids.

material derivative meaning:

$$\vec{a_L} = \frac{d\vec{v}}{dt} + \frac{d\vec{v}}{dx}u + \frac{d\vec{v}}{dy}v + \frac{d\vec{v}}{dz}w$$

$$= \partial\vec{v}/dt + (\vec{v}\cdot\vec{\nabla})\vec{v} = \left(\frac{d}{dt} + (\vec{v}\cdot\vec{\nabla})\right)\vec{v} = \boxed{\frac{D}{Dt}\vec{v}}$$

Euler's equations

① incomp.
② invisc

① $\boxed{-\vec{\nabla}P - \rho g\hat{k} = \rho\frac{D\vec{v}}{Dt}}$ (derivation from Newton's 2nd law)

$\sim \Sigma F$? $\qquad \sim ma$

② $\frac{D\rho}{Dt} + \rho(\vec{\nabla}\cdot\vec{u}) = 0$ (conservation of mass)

when fluid is incompressible.. $\frac{D\rho}{Dt} = 0$

$\therefore \rho(\vec{\nabla}\cdot\vec{u}) = 0 \quad \therefore \boxed{\vec{\nabla}\cdot\vec{u} = 0}$

Intuition of Euler's

$\frac{D\rho}{Dt}$; when $\rho\uparrow$ ⇓ ; when $\rho\downarrow$ ⇓

$-\rho(\vec{\nabla}\cdot\vec{u})$: $\forall\downarrow$; $\forall\uparrow$

synonymous to $\forall$

Navier - stokes

$$\boxed{\rho\frac{D\vec{v}}{Dt} = -\vec{\nabla}P - \rho g\hat{k} + \mu\nabla^2\vec{u}}$$

$\boxed{\vec{\nabla}\cdot\vec{u} = 0}$ → still assumes incompressible

BUT no longer assumes inviscid

Derivation not necessary

Bernoulli's Equations

$$\frac{1}{2}\rho V_1^2 + \rho g z_1 + P_1 = \frac{1}{2}\rho V_2^2 + \rho g z_2 + P_2$$

① inviscid ② incompressible ③ homogenous ④ steady state ⑤ Adiabatic ⑥ streamline exception of direction normal to radius of curvature.

Heuristic justification

Energy in = Energy out. $\qquad E_{k1} + E_{p1} + W_1 = E_{k2} + E_{p2} + W_2$

$$\frac{1}{2}[\rho \cancel{A_1 V_1 t}]V_1^2 + \rho\cancel{A_1 V_1 t}\,g z_1 + P_1\cancel{A_1 V_1 t} = \frac{1}{2}[\rho\cancel{A_2 V_2 t}]V_2^2 + \rho\cancel{A_2 V_2 t}\,g z_2 + P_2\cancel{A_2 V_2 t}$$

$A_1V_1 = A_2V_2$ //mass conservation

MATH justification

$\rho\frac{D\vec{v}}{Dt} + \vec{\nabla}P + \rho g\hat{k} + \mu\nabla^2\vec{u} = 0$ //Navier stokes

parallel to $\vec{v}$ • perpendicular to $\vec{u} = 0$

$\Rightarrow \int_①^② d\vec{r}\cdot\{\rho[\nabla(\frac{1}{2}u^2) - \cancel{\vec{u}\times(\vec{\nabla}\times\vec{u})}] + \vec{\nabla}P + \rho g\hat{k}\} = 0$

intergration from ①→② along a streamline

$\Rightarrow \int_①^② d\vec{r}\cdot\vec{\nabla}[\frac{1}{2}\rho v^2 + P + \rho g\vec{z}] = 0 \xrightarrow{\text{thus}} \boxed{\frac{1}{2}\rho v^2 + P + \rho g h \Big|_①^② = 0}$

As shown, Bernoulli's holds up across a streamline.

STREAMLINES, PATHLINES, STREAKLINES

streamlines: $\frac{dx}{V_x} = \frac{dy}{V_y} = \frac{dz}{V_z}$

pathlines: "we got history" →

$$\boxed{\begin{array}{ll} dx/dt & V_x[x(t), y(t), z(t), t] \\ dy/dt \Rightarrow & V_y[x(t), y(t), z(t), t] \\ dz/dt & V_z[x(t), y(t), z(t), t] \end{array}}$$

function of t_0

NOT a function of time

streaklines : snapshot of multiple particles at any time

Function of time, NOT a function of t_0

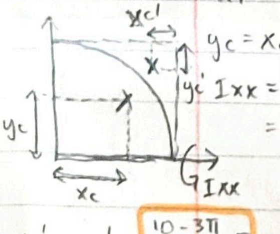
Bernoulli's as applied to a curved flow

$\rho V^2/R = -\frac{\partial}{\partial n}(P - \rho g z) \xrightarrow{\text{intuition}} \rho\frac{V^2}{R} = -\frac{\partial P}{\partial \hat{n}}$

AKA: as you go Along $\hat{n}$ of curved flow, pressure decreases.

↑ P is decreasing

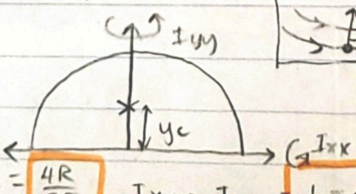
Common Rotational inertias & centroids



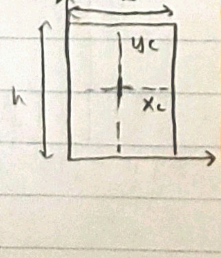
$y_c = x_c = \boxed{\frac{4R}{3\pi}}$

$I_{xx} = \boxed{\frac{1}{16}\pi R^4} = I_{yy}$

$x_c' = y_c' = \boxed{\frac{10 - 3\pi}{12 - 3\pi}R}$



$y_c = \boxed{\frac{4R}{3\pi}} \qquad I_{xx} = I_{yy} = \boxed{\frac{1}{8}\pi R^4}$



$$\boxed{\begin{array}{l} I_x = \frac{bh^3}{3} \\ I_y = hb^3/3 \\ I_{xc} = bh^3/12 \\ I_{yc} = hb^3/12 \end{array}}$$

REMEMBER

$\boxed{y_c = \int y\,dA}$

$\boxed{I_{yy} = \int y^2 dA}$

ME 106 Final Cheatsheet.

Changing Mass Problem

$M(t) = M_0 + \dot{m}t$

CV analysis on [ ]

$\frac{\partial}{\partial t}\int_{cv} \rho \vec{u}\, dA + \int_{cs} \rho \vec{u}(\vec{u}\cdot\hat{n})\, dA$

// Force approach.

$\Rightarrow \frac{\partial}{\partial t}[M(t)v(t)] + \int \rho u(-u)\, dA = 0$

$\Rightarrow \dot{m}v + m\dot{v} - \underbrace{\rho u A}_{\dot{m}} u = 0$

$\Rightarrow \dot{m}v + m\dot{v} = \dot{m}u$

$\Rightarrow \frac{dv}{dt} = \frac{\dot{m}}{m}(u-v) = \frac{\dot{m}(u-v)}{M_0 + \dot{m}t}$

$\int_0^v \frac{dv}{u-v} = \int_0^t \frac{\dot{m}}{M_0 + \dot{m}t} dt$

$\Rightarrow -\ln\left(\frac{u-v}{u}\right) = \ln \frac{M_0 + \dot{m}t}{M_0}$

$\Rightarrow \frac{u}{u-v} = \frac{M_0 + \dot{m}t}{M_0}$

$\boxed{v(t) = u \frac{\dot{m}t}{M_0 + \dot{m}t}}$

Rocket Problem

V_e: exit velo of fuel RELATIVE to rocket.

How to deal with accelerating control volume?

$\frac{\partial}{\partial t}\int_{cv} \cancel{\rho u\, dV} + \int_{cs} \rho u(u\cdot\hat{n})\, dA = \int_{cs} \rho u_e(u_e\cdot\hat{n})\, dA$

FRAME of REFERENCE FOLLOWS THE ROCKET!

$= \int \rho(-V_e^2)\, dA = \underbrace{\Sigma F_{ext}}_{②} - \underbrace{\int_{cv} a\rho\, dV}_{①}$

This term accounts for the fictitious force that comes with using F.O.R. following rocket.

$\Rightarrow -\dot{m}V_e = \underbrace{-mg}_{①} - \underbrace{m\dot{u}}_{②}$ $\xrightarrow[\text{1st order}]{\text{solve}}$ $\boxed{u = -V_e \ln\left(\frac{M_0 - \dot{m}t}{M_0}\right) - gt}$

DIMENSIONAL ANALYSIS

Algorithm:

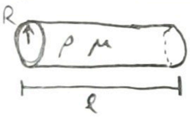
1. Identify all variables that may be potentially relevant to the test at hand. (n)
2. construct table of dimensions (matrix)
3. Find rank of the matrix. (r)
4. Buckingham Pi Thm: There exist $m = n - r$ dimensionless groups in this test/problem
5. Find expressions for $\Pi_1, \Pi_2, \Pi_3 \dots$ set $F(\Pi_1, \dots) = 0$

Tip: $[\mu] = \frac{kg}{ms} = \frac{M}{LT}$

Discussion Alternative Algorithm:

1. List variables $u_1, u_2, \dots u_k$.
2. Determine r (how many fundamental units are relevant here?) Ex; M, L, T $r = 3$.
3. write each fundamental unit (M, L, T) in terms of $u_1 \dots u_k$. (make sure you keep 3 constant choices of $u_1 \dots u_k$)
   - DO NOT CHOOSE A CORE VARIABLE WHEN DEF. FUNDAMENTAL UNITS, THIS LEADS TO THIS VARIABLE POTENTIALLY BEING PRESENT IN EVERY PI → UNECESSARY ALGEBRA.
5. Let $\Pi_i = u_i/[u_i] = u_i / f(u_1, u_2, u_3)$ } from ④

PIPE FLOW EX



$\Pi_1 = \frac{P}{\rho u^2}$

$\Pi_2 = \ell/R$

$\Pi_3 = \frac{\mu}{\rho V R}$

Notice how all are dimensionless

Πs can vary depending on variable choice.

$Re = 1/\Pi_3 = \boxed{\frac{\rho u D}{\mu}}$ Reynold's # ! → use D for pipe flow

↳ use ℓ for flat, planar flow

$Re_{critical} = 2000$ if $Re < Re^{cr}$: laminar flow

if $Re > Re^{cr}$: turbulent flow

Friction Factor

Review of Poiseulle flow:

LAMINAR pipe flow

sanity check!

$\boxed{Q = \bar{u}\pi R^2 = \frac{\pi R^4}{8\mu}\left(\frac{\Delta P}{\Delta x}\right) = \frac{\pi D^4}{128\mu}\left(\frac{\Delta P}{\Delta x}\right)}$

$\bar{u}\cancel{\pi R^2} = \frac{\cancel{\pi} R^{\cancel{4}\,2}}{8\mu}\left(\frac{\Delta P}{\Delta x}\right) \Rightarrow 8 = \frac{R^2}{\mu \ell} * \frac{\Delta P}{u} * \frac{u}{u} * \frac{\rho}{\rho} \Rightarrow \frac{-\Delta P}{\rho u^2} * \frac{\rho u R}{\mu} * \frac{R}{\ell}$

$\Pi_1 * \Pi_2^{-1} * \Pi_3^{-1}$!

$8 = \frac{\Pi_1}{\Pi_2 \Pi_3} = \left(\frac{\Delta P}{\rho u^2}\right)\left(\frac{D}{2\ell}\right)\left(\frac{Re}{2}\right) \Rightarrow \boxed{\frac{64}{Re} = \frac{\Delta P}{\frac{1}{2}\rho u^2} * \frac{D}{\ell} = f}$ FOR LAMINAR FLOW

// couple confusing factors of ½

$\boxed{\Delta P = f \frac{\ell}{D} \frac{\rho V^2}{2}}$ it follows

$\boxed{f \to \text{LOOK in MOODY TABLE for turbulent flow}}$

Intuition for Reynold's #, f, Fr

Reynolds: ratio of INERTIAL / VISCOUS forces.

→ if Re↑, Inertia ↑, μ↓, f↓, less stabilizing viscosity TURBULENCE!

→ if Re↓, Inertia ↓, μ↑, f↑, more stabilizing visc. LAMINAR

Froude #: ratio of INERTIAL/GRAVITATIONAL forces → useful for wave analysis and when fluid weight is non-negligible.

Energy of Pipe flow + Head Loss

$\boxed{\frac{P_1}{\gamma} + \alpha_1\frac{V_1^2}{2g} + z_1 = \frac{P_2}{\gamma} + \alpha_2\frac{V_2^2}{2g} + z_2 + h_L}$

α_1, α_2 (α_1, α_2) ⇒ typically 1 used to acct. for using $\bar{u}$

// all components are in unit LENGTH!

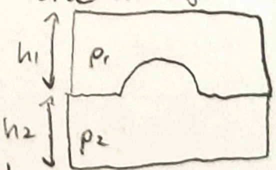
$\boxed{\gamma \Rightarrow \rho g}$ ☆

$\boxed{h_{L\,major} = f \frac{\ell}{D}\frac{V^2}{2g}}$ caused by friction within the pipe

$\boxed{h_{L\,minor} = K_L \frac{V^2}{2g}}$ caused by pipe components like valves, ducts, bends

Buoyancy

Force always acts UPWARDS

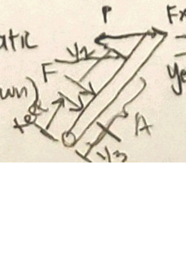


HW 4 Linear Trapdoors

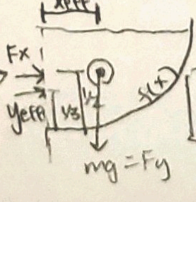
$F = P_{effective} A$

$= \rho g z_{centroid} A$

$\boxed{z_{eff} = \frac{2}{3}z}$ for hydrostatic (AKA 2/3 of the way down)



Odd Trapdoors



Force and moment balance

$\boxed{x_{eff} = \frac{\int x\, dA}{\int dA}}$

COMPRESSIBLE FLOWS — cylinder example

$Re < 1$ — smooth flow as modeled in class

$Re < 40$

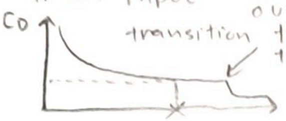
Skin Drag: Drag due to no-slip viscous boundary

Form Drag: Drag due to low pressure ☆

$Re > 40$ — pure chaos

experimental proof Navier stokes is unsolvable: linear input → nonlinear output

$$\boxed{C_D = \frac{F_D}{\frac{1}{2}\rho u^2 DL}} = f(Re)$$



Supersonic flows (through dA pipes)

cons of mass: $\boxed{\frac{d\rho}{\rho} + \frac{dA}{A} + \frac{du}{u} = 0}$

cons of momentum: $\boxed{\frac{dP}{\rho u^2} = -\frac{du}{u}}$ ⑥

combined: $\boxed{\frac{du}{u} = -\frac{dA}{A}\left(\frac{1}{1-Ma^2}\right)}$ ☆

$$\boxed{\frac{d\rho}{\rho} = \frac{dA}{A}\left(\frac{Ma^2}{1-Ma^2}\right)}$$ ☆

	subsonic (Ma<1)	supersonic (Ma>1)	
Diverging duct (dA +)	(du −) (dρ +)	(du +) (dρ −) speed increases!	cons of energy is upheld since ρ has changed
Converging duct (dA −)	du + dρ −	(du −) (dρ +) speed decreases!	

THERMODYNAMICS + the speed of sound

$\boxed{c = \sqrt{\frac{E_v}{\rho}}}$ in fluids where E_v → Bulk Modulus, ρ → fluid density

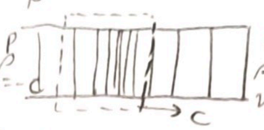
$\boxed{c = \sqrt{KRT}}$ in gases where K → specific heat ratio $\boxed{C_p/C_v}$, R → ideal gas const $\boxed{C_p - C_v}$

$\boxed{\text{Mach \#} = u / C_{@\,altitude}}$ Keep in Mind Mach # denom changes with altitude.

Derivation of the speed of sound:

① isentropic assumed (adiabatic + reversible)

$\frac{P}{\rho^K}$ → constant



② CV moves WITH sound propagation

③ Mass conservation yields $\boxed{\rho\,\delta V = c\,\delta\rho}$

④ Momentum conservation yields $\boxed{\rho\,\delta V = \frac{\delta P}{c}}$

⑤ $c\,\delta\rho = \frac{\delta P}{c} \Rightarrow c^2 = \frac{\delta P}{\delta \rho} \Rightarrow \boxed{c = \sqrt{\frac{\delta P}{\delta \rho}}}$

$$= \sqrt{\frac{\partial}{\partial\rho}(\alpha\rho^k)} = \sqrt{\alpha K \rho^{k-1}} = \sqrt{\frac{P}{\rho^k} K \rho^{k-1}} = \sqrt{\frac{P}{\rho}k}$$

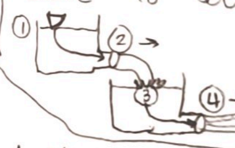
$$= \boxed{\sqrt{KRT}}$$ ☆

HW5

let streamline $= \vec{s}$ — $\vec{s}$ and $\vec{u}$ must be parallel everywhere... $\vec{s} \times \vec{u} = \vec{0}$!

//derivation for streamline justification

Bernoulli's: Beware of discontinuities! best to set up checkpoints!



①→②

③→④

②→④ mass cons

$P_2 = P_4 = P_{atm}$!

HW6

Recall: $\Sigma\phi = \phi_1 + \phi_2 \ldots$, $\Sigma\psi = \psi_1 + \psi_2 + \ldots$ } both are superimposable

to find stagnation points, find $\vec{u}$ using $\Sigma\psi$ or $\Sigma\phi$, set the PROPER components of u to 0 — takes some thinking

cylindrical ⟺ cartesian coordinates.

$$\begin{Bmatrix} \partial\phi/\partial x \\ \partial\phi/\partial y \end{Bmatrix} \leftrightarrow \begin{Bmatrix} \partial\phi/\partial r \\ \frac{1}{r}\,\partial\phi/\partial\theta \end{Bmatrix} \to 0$$

METHOD of IMAGES

HW7

Understand Navier-Stokes

HW8

→ Coutte Flow with nonhomogeneous fluid! → use Coutte Flow of 2 homogenous fluids with 1 unknown boundary condition. set boundaries equal

→ derivation of Taylor-Coutte Flow

$\frac{\partial}{\partial t}\int \rho\, d\forall \to 0$ when system is in steady-state

For streams

$$\int_{CS} \rho \begin{Bmatrix} u_x \\ u_y \end{Bmatrix} \left(\begin{Bmatrix} v_x \\ u_y \end{Bmatrix} \cdot \begin{Bmatrix} A\sin\theta \\ A\cos\theta \end{Bmatrix} \right)$$

you may only have the sin/cos dA term ONCE! No $\sin^2\theta$, $\cos^2\theta$

HW9

$$\boxed{\eta = \frac{\dot{w} - loss}{\dot{w}}}$$

If Pressure remains constant internal E $\tilde{u}$ will cancel out!

since $\tilde{u} = P/\rho$

HW10

Dimensions again... wtf.

Changing Mass Rocket Booster Problem!

frame of reference with the rocket

$\vec{u} = 0$, $\vec{u}_{fuel} = u_{exit}$! $M\vec{a}$ = fictitious force

$\dot{\vec{u}} = \vec{a}$

how easy is that?!

$$\cancel{\frac{\partial}{\partial t}\int \rho \vec{u}\, d\forall}^{\,0} + \int_{CS} \rho \vec{u}(\vec{u}\cdot\hat{n})\, dA = -M\vec{g} - M\vec{a}$$

$$\rho_{fuel}\, \vec{u}_{fuel}^{\,2} A = \text{" \quad "}$$

$$\dot{M} u_{fuel} = -Mg - Ma^{\to \dot{u}}$$

$$= -Mg - M\dot{u}$$

//solve first order DEQ
~~plug in~~ $M = M_0 + \dot{M}t$ ☆

HW11

Pipe Flows, Turbulence, compressibles.

if flow is turbulent $\boxed{\frac{l_e}{D} = 4.4\, Re^{1/6}}$

laminar $\boxed{\frac{l_e}{D} = 0.06\, Re}$

l_e: entrance length (distance fluid must travel before being fully developed)

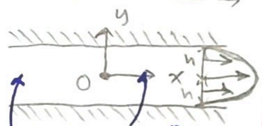
Pressure ratio

$$\boxed{\frac{P_0}{P} = \left(1 + \frac{V^2}{2C_pT}\right)^{K/(K-1)}} \quad \boxed{K = \frac{c_p}{c_v}}$$

K and R vary by material/temperature!

Navier-Stokes derivations!

Couette Flow
→ steady $\frac{d}{dt}=0$
→ laminar $v=w=0$



$\vec{u}=\begin{Bmatrix} u \\ v \end{Bmatrix}$ $u(y=+h)=0$ $u(y=-h)=0$ // Boundary conditions

higher P lower P
$\frac{\partial p}{\partial x}$ negative!

$$\boxed{u=\frac{y^2}{2\mu}\left(\frac{\partial p}{\partial x}\right)+C_1y+C_2}$$

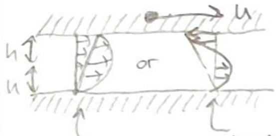
$C_1=0$
$C_2=-\frac{1}{2\mu}\left(\frac{\partial p}{\partial x}\right)h^2$

// solve for C_1 and C_2 using boundary conditions.

$$\boxed{Q=\int u\,dA=\frac{2b}{3\mu}\left(-\frac{\partial p}{\partial x}\right)h^3}$$

when $\frac{\partial p}{\partial x}>0$, $u,Q<0$
$\frac{\partial p}{\partial x}<0$, $u,Q>0$

Couette Flow w/moving plate

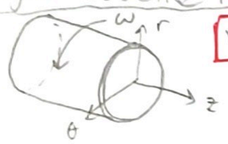


when $u>0$ when $u<0$

$$\boxed{u=\frac{1}{2\mu}\left(\frac{\partial p}{\partial x}\right)(y^2-h^2)+\frac{u(y+h)}{2h}}$$

$u(y=-h)=0$ $u(y=h)=u$
new boundary conditions for new c_1, c_2

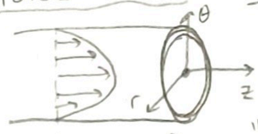
Taylor-Couette Flow



$$\boxed{v_\theta(r)=C_1r+\frac{C_2}{r}}$$

→ $C_2=0$ when cylinder is empty
↳ $C_2\neq0$ when there is an embedded cylinder

Poiseuille Flow
→ steady $\frac{d}{dt}=0$
→ laminar $v=w=0$



$u=\begin{Bmatrix} \cancel{u_r}\;0 \\ \cancel{u_\theta}\;0 \\ u_z \end{Bmatrix}$ $\frac{\partial u_z}{\partial z}=0!$ // cons of mass

Equations from "the" sheet

① $0=-\rho g\sin\theta-\frac{\partial p}{\partial r}$
② $0=-\rho g\cos\theta-\frac{1}{r}\frac{\partial p}{\partial\theta}$
③ $0=-\frac{\partial p}{\partial z}+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right)\right]$

$u_z=\frac{1}{4\mu}\left(\frac{\partial p}{\partial z}\right)r^2+c_1\ln(r)+c_2$

$c_1=0$ $c_2=-\frac{1}{4\mu}\left(\frac{\partial p}{\partial z}\right)R^2$

$$\boxed{u_z=\frac{1}{4\mu}\left(\frac{\partial p}{\partial z}\right)(r^2-R^2)}\quad\boxed{Q=\int u_z\,dA=\frac{\pi R^4\Delta p}{8\mu l}}$$

Poiseulle Flow under g

$$\boxed{\begin{aligned}u_z&=\frac{1}{4\mu}\left(\frac{\partial p}{\partial z}-\rho g\right)(r^2-R^2)\\ Q&=\frac{\pi R^4}{8\mu}\left(-\frac{\partial p}{\partial z}+\rho g\right)\end{aligned}}$$

// sanity check! in the presence of ρg, does u_z/Q flow faster or slower

Flow through Annulus

$$\boxed{\begin{aligned}u_z&=\frac{1}{4\mu}\left(\frac{\partial p}{\partial z}\right)\left[r^2-r_o^2+\frac{r_i^2-r_o^2}{\ln(r_o/r_i)}\ln\left(\frac{r}{r_o}\right)\right]\\ Q&=-\frac{\pi}{8\mu}\left(\frac{\partial p}{\partial z}\right)\left[r_o^4-r_i^4-\frac{(r_o^2-r_i^2)^2}{\ln(r_o/r_i)}\right]\end{aligned}}$$

Control-Volume Analysis.

$$\boxed{\frac{DB_{cm}}{Dt}=\frac{\partial}{\partial t}\int_{cv}\rho b\,d\forall+\int_{cs}b\rho(\vec{u}\cdot\hat{n})\,dA}$$

→ B is extensive (mass-dep)
b is intensive (mass-indep)

for a moving frame of reference, replace with $(\underbrace{\vec{u}-\vec{v}_{cs}}_{\text{relative velocity}})$

conceptual basis.
- control volume (arbitrary enclosure)
- control surface (separates universe from cv)
- control mass (closed system, material vol)

RTT

"what's already there + what goes in - what goes out = what's there now"

$$\frac{D}{Dt}B_{cm}=\frac{\partial}{\partial t}B_{cv}-\dot{B}_{cv}^{in}+\dot{B}_{cv}^{out}=\frac{B_{cm}(t+\delta t)-B_{cm}(t)}{\delta t}$$

schools of thought

① mass: let $B_{cm}=m$ $b=1$

$\frac{DB_m}{Dt}=\frac{\partial}{\partial s}\int_{cv}\rho\,d\forall+\int_{cs}\rho(u\cdot\hat{n})\,dA$

shortcuts: $\boxed{v_1A_1=v_2A_2}$

② Force: let $B_{cm}=mu$ $b=u$

$\frac{DB_{cm}}{Dt}=m\dot{u}=ma=F_{ext}$

$\frac{Dmu}{Dt}=\frac{\partial}{\partial t}\int_{cv}\rho u\,d\forall+\int_{cs}\rho\vec{u}(\vec{u}\cdot\hat{n})\,dA$

$\boxed{F=\rho v^2A}$

③ Energy: let $B_{cm}=E$
$B_{cm}=\frac{1}{2}mv^2+mgh+m\tilde{u}$ ← internal E
$b=\frac{1}{2}u^2+gh+\tilde{u}$

$\boxed{\tilde{u}_1=P_1/\rho_1}$ // for internal energy

Move on the Rankine oval:

$\phi=Ur\cos\theta-\frac{m}{2\pi}(\ln r_1-\ln r_2)$
$\psi=Ur\sin\theta-\frac{m}{2\pi}(\theta_1-\theta_2)$
$x_{stag}=\pm\sqrt{a^2+\frac{ma}{\pi U}}$
$h=\frac{a}{2}\left[\left(\frac{h}{a}\right)^2-1\right]\tan\left[\frac{2h}{a}\cdot\frac{\pi Ua}{m}\right]$

HW1 Tricky stuff:

Taylor exp: $f(x)=f(a)+f'(a)(x-a)+f''(a)\frac{(x-a)^2}{2!}+\dots+f^n(a)\frac{(x-a)^n}{n!}$

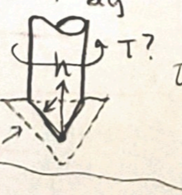
$\vec{u}\cdot\nabla\vec{u}=(\vec{u}\cdot\nabla)\vec{u}$
// remember that $\nabla\vec{u}$ is a VECTOR $\left[\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial z}\right]\vec{u}$

curl: $\nabla\times\vec{u}$
div: $\nabla\cdot\vec{u}$
$\text{div}(\text{curl}(\vec{u}))=0!$

Divergence Thm.
$\oiint\vec{F}\cdot d\vec{s}=\iiint\text{div}\,\vec{F}\,d\forall$
Stokes:
$\oint\vec{F}\cdot d\vec{r}=\iint(\nabla\times\vec{F})\,d\vec{S}$

HW2

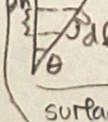
$\tau=\mu\frac{du}{dy}$!



$\boxed{T=\int\tau\times r\,dA}$

$\tau(h)=\mu\frac{du}{dy}$
$=\frac{\mu}{y}u(h)$
$=\frac{\mu}{y}(\omega*r(h))$!

$dA=\underbrace{2\pi r(h)}_{\text{circumference}}\,dl$!



$dl=\frac{dh}{\cos\theta}$

surface tension problems: boundary force holds up the weight of fluid.

HW3 Rotating U-tube

find Pressure as a function of (r, z)
$\vec{\nabla}P=\rho\vec{a}+\rho g\hat{z}=-\rho w^2r\hat{r}-\rho g\hat{z}$

$P=(r,z)=\int-\rho w^2r\,dr+\int-\rho g\,dz$
$=-\frac{1}{2}\rho w^2r^2-\rho gz+C$

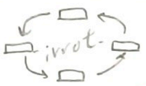
3 unknowns: $C_{mystery}$, C_{water}, P_{water}.
Use boundary conditions
$P_{unknown}(r_{surf},z_{surf})=P_{atm}$
$P_{water}(r_{surf},z_{surf})=P_{atm}$
$P_{unk}(\text{contact point})=P_{water}(\text{contact point})$

Surface Angle Problems:
$P_{surf}=P_{atm}\rightarrow$ const
find $\frac{dy}{dx}$ of surf. set $\frac{\partial p}{\partial x}dx=-\frac{\partial p}{\partial y}dy$ @ surf.

SOLVE $\boxed{\frac{dy}{dx}=\frac{\partial p}{\partial x}\Big/-\frac{\partial p}{\partial y}=-\rho ax/-\rho gy}$

106 Midterm 2 Cheatsheet

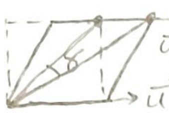
Irrotationality



Vorticity $\boxed{\vec{\omega} = \vec{\nabla} \times \vec{u}}$

Curl of the velocity!

Deformation of fluid blocks



$\delta\gamma \rightarrow$ (negative)

$$\boxed{\frac{\delta\gamma_u}{\partial t} = \ominus\frac{1}{2}\frac{\partial u}{\partial y}}$$

$\delta\gamma \rightarrow$ (positive)

$$\boxed{\frac{\delta\gamma_u}{\partial t} = \frac{1}{2}\frac{\partial u}{\partial t}}$$

understand the sign change and HOW γ is defined!!

combine 2 equations

$$\frac{\partial\gamma}{\partial t} = \frac{1}{2}\left(\frac{\partial u}{\partial t} - \frac{\partial u}{\partial t}\right) = \frac{1}{2}(\vec{\nabla}\times\vec{u})$$

For irrotationality, $\frac{\partial\gamma}{\partial t} = 0 = \frac{1}{2}\vec{\omega} \Rightarrow \vec{\omega} = \boxed{(\vec{\nabla}\times\vec{u}) = 0}$

condition for irrot.

$\vec{\nabla}\times\vec{\nabla}\phi = 0$ where ϕ is some vector

//math property! curl(div($\vec{v}$)) = 0

$\boxed{\vec{u} = \vec{\nabla}\phi}$ only holds, when IRROT

Velocity Potentials

$\vec{\nabla}\times\vec{u} = \vec{\nabla}\times\vec{\nabla}\phi$

$= \boxed{\vec{\nabla}^2\phi = 0}$

assume irrot. and incomp.

Type	ϕ - function	ψ - function (stream)
uniform	$\phi = Ux\cos\theta + Uy\sin\theta$	$\psi = Uy\cos\theta + Ux\sin\theta$
source/sink	$\phi = \pm\frac{m}{2\pi}\ln(r)$	$\psi = \pm\frac{m}{2\pi}\theta$
Forced vortex	$\phi = X$ nonexistent because NOT irrot.	$\psi = X$
Free Vortex	$\phi = k\theta$	$\psi = -k\ln(r)$
Doublet	$\phi = \frac{K\cos\theta}{r}$	$\psi = -\frac{K}{r}\sin\theta$

Relationship between $\vec{u}, \phi, \psi$

$\vec{u} = \nabla\phi = \boxed{\begin{Bmatrix}\partial/\partial x \\ \partial/\partial y\end{Bmatrix}\begin{Bmatrix}\phi_u \\ \phi_u\end{Bmatrix}}$

$\vec{u} = \begin{Bmatrix}u \\ u\end{Bmatrix}$ $\boxed{u = \frac{\partial\psi}{\partial y}}$ $\boxed{u = -\frac{\partial\psi}{\partial x}}$

Special property of ψ

$\partial\psi = u\partial y - u\partial x$

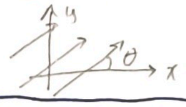
when ψ = const, $\partial\psi = 0$

$0 = u\partial y - u\partial x \Rightarrow \boxed{\frac{\partial y}{\partial x} = \frac{u}{U}}$

streamline equation!

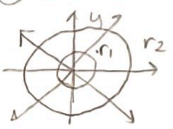
ψ is constant along streamline!

(1) Uniform



pay attention to how θ is defined!

(2) Source/Sink — Rigorous proof



$u_1(\cancel{2\pi}r_1) = u_2(\cancel{2\pi}r_2)$ //cons of mass

$V(r) = \frac{m}{2\pi r}$ this holds true

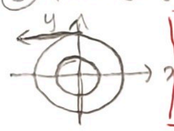
$\begin{Bmatrix} m/2\pi r \\ 0 \\ 0 \end{Bmatrix} = \begin{Bmatrix} \partial\phi/\partial r \\ \partial\phi/\partial\theta * \frac{1}{r} \\ \partial\phi/\partial z \end{Bmatrix}$ m is strength!

$\frac{\partial\phi}{\partial r} = \frac{m}{2\pi r}$ $\int\partial\phi = \int\frac{m}{2\pi r}\partial r$ $\boxed{\phi = \frac{m}{2\pi}\ln(r)} + C(\theta, z)$

$\frac{\partial\phi}{\partial\theta} * \frac{1}{r} = 0$ $\frac{\partial\phi}{\partial\theta} = 0$ ϕ is NOT a function of θ

$\frac{\partial\phi}{\partial z} = 0$ ϕ is NOT a function of z

(3) Forced Vortex.



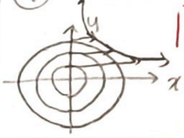
$\boxed{\begin{aligned}V_\theta(r) &= \omega r \\ V_r &= 0 \\ V_z &= 0\end{aligned}}$

$\vec{\nabla}\times\vec{u} = \begin{Bmatrix}0 \\ 0 \\ 2\omega\end{Bmatrix} = \vec{0}$!

rotational flow!

directly proportional to r

(4) Free Vortex



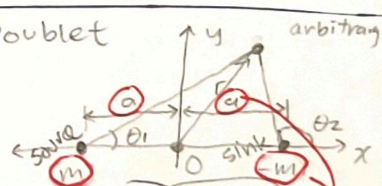
inversely proportional to r

$\boxed{u_\theta = \frac{K}{r}}$ where $\boxed{K = \frac{\Gamma}{2\pi}}$

$\phi = k\theta$

$\psi = -k\ln(r)$

(5) Doublet



For rigorous derivation. set $\boxed{\lim_{\frac{a}{r}\to 0}}$!

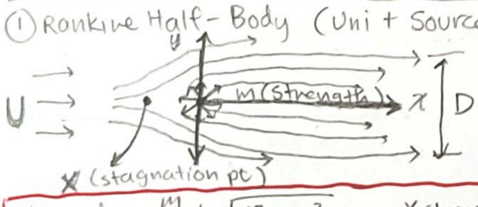
$\psi = -\frac{ma}{\pi r}\sin\theta$

$\phi = \frac{ma}{\pi r}\cos\theta$

$\boxed{K = \frac{ma}{\pi}}$

Common Potential Superposition Combos

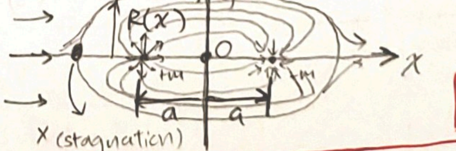
(1) Rankine Half-Body (Uni + Source)



$$\boxed{\begin{aligned}\phi &= Ux + \frac{m}{2\pi}\ln\sqrt{x^2+y^2} & X_{stagnation} &= -\frac{m}{2\pi U} \\ &= Ur\cos\theta + \frac{m}{2\pi}\ln(r) & D_{(width\ as\ x\to\infty)} &= \frac{m}{U}\end{aligned}}$$

cheat codes!!

(2) Rankine Oval (Uni + Source + Sink)

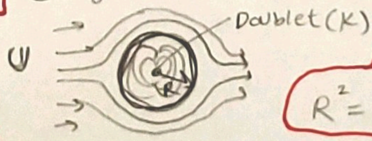


X stagnation: solve x in the following:

$$\boxed{(x^2-a^2)^2 = \left(\frac{m}{4\pi U}\right)4ax}$$

$$\boxed{\phi = Ux + \frac{m}{2\pi}\left(\ln\sqrt{(x+a)^2+y^2} - \ln\sqrt{(x-a)^2+y^2}\right)}$$

(3) Cylindrical Flows (Uni + Doublet)

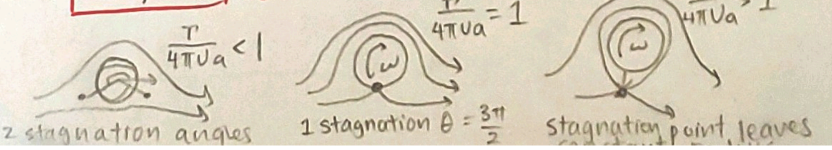


$$\boxed{\begin{aligned}\phi &= Ur\cos\theta + \frac{K\cos\theta}{r} \\ \psi &= Ur\sin\theta - \frac{K\sin\theta}{r} \\ R^2 &= K/U\end{aligned}}$$

(4) Rotating Cylinders (Uni + Doublet + Vortex)

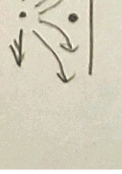
$$\boxed{\begin{aligned}\phi &= Ur\cos\theta + \frac{K\cos\theta}{r} + \frac{\Gamma}{2\pi}\theta \\ \psi &= Ur\sin\theta - \frac{K\sin\theta}{r} - \frac{\Gamma}{2\pi}\ln(r) \\ \sin\theta_{stag} &= \frac{\Gamma}{4\pi Ua}\end{aligned}}$$

Lift $\boxed{F_L = \rho U\Gamma}$!!



(5) Source against a wall

method of images!



Cartesian Coordinate:

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0 \quad \Big\} \text{ cons of mass}$$

$$\vec{\nabla}\cdot\vec{u}=0$$

$$\text{x-comp } \Big\{ \rho\left[\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right]=-\frac{\partial p}{\partial x}+\mu\left[\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right]+F_x$$

$$\text{y-comp } \Big\{ \rho\left[\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\right]=-\frac{\partial p}{\partial y}+\mu\left[\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\right]+F_y$$

$$\text{z-comp } \Big\{ \rho\left[\frac{\partial w}{\partial t}+u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\right]=-\frac{\partial p}{\partial z}+\mu\left[\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}+\frac{\partial^2 w}{\partial z^2}\right]+F_z$$

Cylindrical Coordinate:

$$\frac{\partial u_r}{\partial r}+\frac{u_r}{r}+\frac{1}{r}\frac{\partial u_\phi}{\partial \phi}+\frac{\partial u_z}{\partial z}=0 \quad \Big\} \text{ cons of mass}$$

$$\left[\frac{\partial u_r}{\partial t}+u_r\frac{\partial u_r}{\partial r}+\frac{u_\phi}{r}\frac{\partial u_r}{\partial \phi}+u_z\frac{\partial u_r}{\partial z}-\frac{u_\phi^2}{r}\right]=-\frac{\partial p}{\partial r}+\mu\left[\frac{\partial^2 u_r}{\partial r^2}+\frac{1}{r}\frac{\partial u_r}{\partial r}-\frac{u_r}{r^2}+\frac{1}{r^2}\frac{\partial^2 u_r}{\partial \phi^2}+\frac{\partial^2 u_r}{\partial z^2}-\frac{2}{r^2}\frac{\partial u_\phi}{\partial \phi}\right]+F_r$$

$$\left[\cancelto{0}{\frac{\partial u_\phi}{\partial t}}+\cancelto{0}{u_r\frac{\partial u_\phi}{\partial r}}+\cancelto{0}{\frac{u_r u_\phi}{r}}+\cancelto{0}{\frac{u_\phi}{r}\frac{\partial u_\phi}{\partial \phi}}+\cancelto{0}{u_z\frac{\partial u_\phi}{\partial z}}\right]=-\frac{1}{r}\frac{\partial p}{\partial \phi}+\mu\left[\frac{\partial^2 u_\phi}{\partial r^2}+\frac{1}{r}\frac{\partial u_\phi}{\partial r}-\frac{u_\phi}{r^2}+\cancelto{0}{\frac{1}{r^2}\frac{\partial^2 u_\phi}{\partial \phi^2}}+\cancelto{0}{\frac{\partial^2 u_\phi}{\partial z^2}}+\cancelto{0}{\frac{2}{r^2}\frac{\partial u_r}{\partial \phi}}\right]+F_\phi$$

$$\left[\frac{\partial u_z}{\partial t}+u_r\frac{\partial u_z}{\partial r}+\frac{u_\phi}{r}\frac{\partial u_z}{\partial \phi}+u_z\frac{\partial u_z}{\partial z}\right]=-\frac{\partial p}{\partial z}+\mu\left[\frac{\partial^2 u_z}{\partial r^2}+\frac{1}{r}\frac{\partial u_z}{\partial r}+\frac{1}{r^2}\frac{\partial^2 u_z}{\partial \phi^2}+\frac{\partial^2 u_z}{\partial z^2}\right]+F_z$$

Spherical Coordinate:

will not use as much in class

$$\frac{\partial u_r}{\partial r}+\frac{2u_r}{r}+\frac{1}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_\theta \cot\theta}{r}+\frac{1}{r\sin\theta}\frac{\partial u_\phi}{\partial \phi}=0$$

$$\rho\left[\frac{\partial u_r}{\partial t}+u_r\frac{\partial u_r}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta}+\frac{u_\phi}{r\sin\theta}\frac{\partial u_r}{\partial \phi}-\frac{u_\theta^2}{r}-\frac{u_\phi^2}{r}\right]=-\frac{\partial p}{\partial r}$$
$$+\mu\left[\frac{\partial^2 u_r}{\partial r^2}+\frac{2}{r}\frac{\partial u_r}{\partial r}-\frac{2u_r}{r^2}+\frac{1}{r^2}\frac{\partial^2 u_r}{\partial \theta^2}+\frac{\cot\theta}{r^2}\frac{\partial u_r}{\partial \theta}+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 u_r}{\partial \phi^2}-\frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta}-\frac{2u_\theta\cot\theta}{r^2}-\frac{2}{r^2\sin\theta}\frac{\partial u_\phi}{\partial \phi}\right]+F_r$$

$$\rho\left[\frac{\partial u_\theta}{\partial t}+u_r\frac{\partial u_\theta}{\partial r}+\frac{u_r u_\theta}{r}+\frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_\phi}{r\sin\theta}\frac{\partial u_\theta}{\partial \phi}-\frac{u_\phi^2\cot\theta}{r}\right]=-\frac{1}{r}\frac{\partial p}{\partial \theta}$$
$$+\mu\left[\frac{\partial^2 u_\theta}{\partial r^2}+\frac{2}{r}\frac{\partial u_\theta}{\partial r}-\frac{u_\theta}{r^2\sin^2\theta}+\frac{1}{r^2}\frac{\partial^2 u_\theta}{\partial \theta^2}+\frac{\cot\theta}{r^2}\frac{\partial u_\theta}{\partial \theta}+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 u_\theta}{\partial \phi^2}+\frac{2}{r^2}\frac{\partial u_r}{\partial \theta}-\frac{2\cot\theta}{r^2\sin\theta}\frac{\partial u_\phi}{\partial \phi}\right]+F_\theta$$

$$\rho\left[\frac{\partial u_\phi}{\partial t}+u_r\frac{\partial u_\phi}{\partial r}+\frac{u_r u_\phi}{r}+\frac{u_\theta}{r}\frac{\partial u_\phi}{\partial \theta}+\frac{u_\theta u_\phi\cot\theta}{r}+\frac{u_\phi}{r\sin\theta}\frac{\partial u_\phi}{\partial \phi}\right]=-\frac{1}{r\sin\theta}\frac{\partial p}{\partial \phi}$$
$$+\mu\left[\frac{\partial^2 u_\phi}{\partial r^2}+\frac{2}{r}\frac{\partial u_\phi}{\partial r}-\frac{u_\phi}{r^2\sin^2\theta}+\frac{1}{r^2}\frac{\partial^2 u_\phi}{\partial \theta^2}+\frac{\cot\theta}{r^2}\frac{\partial u_\phi}{\partial \theta}+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 u_\phi}{\partial \phi^2}+\frac{2}{r^2\sin\theta}\frac{\partial u_r}{\partial \phi}+\frac{2\cot\theta}{r^2\sin\theta}\frac{\partial u_\theta}{\partial \phi}\right]+F_\phi$$